



BIRLA INSTITUTE OF TECHNOLOGY AND SCIENCE, Pilani
Pilani Campus
AUGS/ AGSR Division

FIRST SEMESTER 2026-27
COURSE HANDOUT

Date: 01.08.2026

In addition to part I (General Handout for all courses appended to the Time table) this portion gives further specific details regarding the course.

Course No : ME G532
Course Title : Machine Tool Engineering
Instructor-in-Charge : Prof. Anuj Sharma [anuj.sharma@pilani.bits-pilani.ac.in , 1215-N]

1. Course Description: Design principles of machine tools; stiffness and rigidity of separate construction elements and their combined behaviour under load; design of stepped and stepless drives; electrical, mechanical and hydraulic drives; design of bearings and slideways; machine tool controls; machine tool dynamics; recent developments in machine tool design

2. Scope and Objective of the Course: The Course is intended to focus the student's attention on the concepts of design, analysis and features of different types of machine Tools, different machine elements to make a machine tool and new concepts in machine tool engineering.

Course Objectives (CO):

- CLO1. To advance the theoretical and fundamental understanding of machining processes
- CLO2. To understand the concept of hydraulic and mechanical transmissions employed in machine tools
- CLO3. Calculation of machining time of various operations
- CLO4. To understand the design of gear boxes for stepped control of speed and feed in machine tools
- CLO5. To understand the design solution of different machine tool elements
- CLO6. To understand the functioning and operation of numerically control machine tools.

3. Text Books:

TB1. NK Mehta," Machine Tool Design & Numerical Control", third edition, Tata McGraw Hill Pvt. Ltd, New Delhi, 2012

References books:

- RB1. G.C Sen and A. Bhattacharya, "Principles of Machine Tools", New Central Book agency, Calcutta, 1998
- RB2. A.B Chattopadhyay, "Machining and Machine Tools" Wiley-India, 2011

4. Course Prerequisites:

Theory of metal cutting, Machine design

5. Course Plan:

Week No.	Module No.	Lecture Session	Reference	Learning outcomes
1-3	1. Introduction to machine tool drives and mechanisms	L1-7: Machining Fundamentals, Working and auxiliary motions of a machine tool, Machine tool drives and elements	T1 (Chapter -1) R2 (Chapter-3,7)	Calculation of machining time and concept of hydraulic and mechanical transmissions employed in machine tools



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3-7	2. Regulation of speed and feed rates	L8-20: Various laws of speed regulation, Structural diagrams, Speed charts	T1 (Chapter -2) R1 (Chapter-3)	Design of gear boxes for stepped control of speed and feed in machine tools
7-9	3. Design of machine tool structures	L21-27: Design criteria, material, design procedure of machine tool structures	T1 (Chapter -3)	Design of bed, column, housing, base and table; machine tool stiffness
10	4. Design of guideways and power screws	L28-30: Functions, material, design criteria, design procedure of guideways and powerscrews	T1 (Chapter -4)	Design principle of power screws and guideways
11-12	5. Design of spindle and spindle supports	L31-34: Functions, material, and design calculations of the spindle and supports	T1 (Chapter -5)	Design of spindle and spindle supports
12-14	6. Numerical control of machine tools	L35-40: Structure of numerical control systems, Manual part programming	T1 (Chapter -8)	Functioning and operation of numerically control machine tools

6. Evaluation Scheme:

Component	Duration	Weightage (%)	Date & Time	Nature of component (Close Book/ Open Book)
Mid-Semester Test	90 Min.	25	08/10 AN2	Close book
Comprehensive Examination	3 h	35	12/12 AN	Close book
Assignment: Report writing on literature/Research	-	20	To be announced in class	Open book
Presentation on Process fundamentals/Recent developments/Research	-	20	To be announced in class	Open book

7. Chamber Consultation Hour: Tuesday (5:30-6:30 PM)

8. Criterion for NC: 20% of the average of the top three performers or 30% of median, whichever is lower,

9. Attendance Policy: Each student is expected to attend all classes and participate in all evaluation components.

10. Notices: All notices will be displayed through either Emails or the Nalanda portal.

11. Make-up Policy: Makeup will be given only for genuine cases (accompanied by proof) for Mid-sem



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or comprehensive exam only.

12. Note (if any):



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